

**FOURTH SEMESTER DIPLOMA MODEL PAPER IN COMPUTER
APPLICATION AND BUSINESS MANAGEMENT
RELATIONAL DATABASE MANAGEMENT SYSTEMS**

[Time : 3 hours]

(Maximum Marks: 100)

PART – A
(Maximum Marks: 10)

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define Primary Key.
2. Distinguish between Physical & Logical Records.
3. Interpret Decomposition of the relation.
4. List any four Data types in SQL.
5. Define Cursor.

(5x2=10)

PART – B
(Maximum Marks: 30)

II Answer any five questions. Each question carries 6 marks.

1. Outline the advantages of DBMS.
2. Distinguish the database sub languages – DDL, DML & DCL.
3. Explain the set operation: UNION and INTERSECTION with suitable examples.
4. With appropriate examples compare BCNF & 4 NF.
5. Write the general Syntax of ALTER TABLE Command. Explain any four variants of ALTER TABLE Command with suitable examples
6. Demonstrate the following integrity constraints with examples.
a) DEFAULT b) CHECK c) UNIQUE
7. Define Trigger. Illustrate the purpose of Trigger with suitable example.

(5x6=30)

PART – C
(Maximum Marks: 60)

Answer one full question from each module. Each question carries 15 Marks

MODULE-I

- III a) Explain three schema architecture with a neat sketch. 9
b) State the different types of database users and their privileges. 6

OR

- IV Explain ER Model bringing out the different components of the model. Illustrate with an example. 15

MODULE-II

- V a) State the significance of mapping ER model to Relational Model. Outline different steps involved in the process 9
b) Define 5 NF and explain join dependencies 6

OR

- VI Illustrate functioning of following Relational algebra operations with examples
a) SELECT b) PROJECT c) REANME
d) DIFFERENCE e) CARTESIAN PRODUCT 15

MODULE-III

- VII a) Identify the purpose of Query Optimization. Outline the 3 types of Query Optimization Methods. 9
b) Illustrate the use of the following aggregate functions.
a) SUM () b) AVG () c) COUNT () d) MIN () 6

OR

- VIII Apply SQL commands for following operations. 15
a) Create a relation named STUDENT with following details.
Attributes- RegNo. , Name , Branch, Mark1, Mark2, Mark3. (Primary Key- RegNo.)
b) Write SQL commands to insert 5 set of values in the above relation with minimum of two branches (Computer and Electronics)
c) Write a SQL Command to select details of all students of Computer department.
d) Write a SQL Command to add an attribute 'TOTAL' to the relation.
e) Write a SQL Command to update value of TOTAL as sum of Mark1, Mark2 & Mark3

MODULE-IV

IX a) Compare Embedded SQL and Dynamic SQL outlining the features , advantages & disadvantages of the both 9

b) Elaborate the concept of stored procedure with suitable example 6

OR

X Explain the transaction concepts- Properties of transaction, states of transaction and concurrent execution with required diagrams 15
